

FactoryOps AI – Building an Intelligent Manufacturing Assistant

Your team at **GlobalLogic** has been hired by **FactoryOps Manufacturing** to build an AI-powered Manufacturing Assistant for machine operators, maintenance engineers, production supervisors, and plant managers.

The client doesn't want a finished product immediately. They want an MVP that grows over multiple phases. Your job is to implement those phases while adapting to new business requirements.

The same application evolves across multiple levels.

Client Profile

Organization

FactoryOps Manufacturing

- 5 Manufacturing Plants
- 3,500 Employees
- 250 Production Machines
- 24x7 Production Operations
- Multiple Product Lines

Existing Technology Landscape

- Manufacturing Execution System
- Machine Monitoring Dashboard
- Maintenance Management System
- Inventory Management System
- Quality Inspection Portal
- Email & Teams
- REST APIs for internal systems

Current Business Challenges

Machine Operations

Operators frequently ask questions regarding:

- Machine operating procedures
- Error codes
- Safety checks
- Production targets
- Shift handover instructions

Maintenance

Maintenance teams manually handle:

- Machine fault reporting
- Technician assignment
- Maintenance ticket creation
- Downtime tracking

Production Management

Supervisors spend significant time:

- Checking machine status
- Reviewing downtime
- Coordinating maintenance
- Preparing shift summaries

Executive Goals

The COO expects:

- Reduced machine downtime
- Faster maintenance response
- Better operator productivity
- Improved production visibility
- AI-assisted plant operations

Level 1 : Build an AI Manufacturing Assistant

Machine operators frequently need help with basic operating procedures, machine issues, and safety checks.

The factory wants a conversational AI assistant capable of answering general manufacturing-related questions.

At this stage, there is no machine or maintenance system integration. The assistant behaves like a capable LLM-powered chatbot.

Technical Focus

- LLM Integration
- Prompt Engineering
- Chat Interface
- Conversation Memory
- System Prompts
- Message History
- Basic LangChain Components

Learners Build

- Chat UI
- Conversation History
- Manufacturing Assistant Persona
- Prompt Templates
- Basic Guardrails
- Session Memory

Stretch Challenge

Different users should receive different responses depending on whether they are:

- Machine Operator
- Maintenance Engineer
- Production Supervisor

Level 2 : Transform the Assistant into an AI Agent

The assistant should now perform actions, not just answer manufacturing questions.

Examples:

- Check Machine Status
- Retrieve Error Details
- Create Maintenance Ticket
- Check Technician Availability
- Generate Shift Summary
- Escalate Critical Machine Failure

Technical Focus

- Agents
- Tool Calling
- LangGraph
- MCP Concepts
- Multi-step Reasoning
- Human-in-the-loop

Learners Build

Convert yesterday's chatbot into an Agent.

Create tools like:

- Machine Status Checker
- Error Code Lookup
- Maintenance Ticket Generator
- Technician Availability Checker

The agent should decide which tool or tools to invoke based on the user's request.

Stretch Challenge

If a machine failure is marked as critical, the assistant should ask for supervisor approval before creating a shutdown request.

Level 3 : Simplified Multi-Agent Architecture

A user submits:

“Machine MX-204 has stopped with error code E17. Check the issue and create a maintenance request.”

The system should:

1. Understand the machine issue.
2. Retrieve machine details.
3. Check the meaning of the error code.
4. Determine the issue severity.

5. Recommend the required maintenance action.
6. Ask the user to confirm.
7. Create a simulated maintenance request.

This is intentionally a **machine fault-handling workflow**, not a complete predictive-maintenance platform.

1. Fault Analysis Agent

Responsibilities

- Extract machine ID
- Extract error code
- Identify the requested action
- Determine whether information is missing

Structured Output

None

```
• {  
•   "machine_id": "MX-204",  
•   "error_code": "E17",  
•   "request_type": "Maintenance Request",  
•   "missing_fields": []  
}
```

2. Maintenance Diagnosis Agent

Responsibilities

- Search machine information from a provided JSON file
- Check error details from another JSON file
- Determine issue severity
- Recommend the required maintenance action

Tools

None

- `search_machine()`
- `lookup_error_code()`

This agent combines machine and error analysis to keep the lab manageable.

3. Maintenance Request Agent

Responsibilities

- Present the recommended action
- Pause for user approval
- Create a simulated maintenance request
- Generate confirmation

Tool

None

`create_maintenance_ticket()`

The tool can append the ticket to an in-memory list or local JSON file.

Level 4 : Deploy Your AI Application

The client is satisfied with the application and wants it to be accessible to internal users for demonstration and testing.

Your task is to deploy the AI application and make it available through a web interface.

Technical Focus

- Streamlit Deployment
- Environment Variables

- Configuration Management
 - Application Packaging
 - Basic Logging
 - Production Readiness
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Learners Build

- Configure environment variables
- Prepare a production-ready `.env`
- Organize project structure
- Deploy the application using **Streamlit**
- Verify that all previous levels function correctly after deployment

Learning Outcome

Learners complete all three levels by progressively building a real-world AI application.

They understand:

- Prompt Engineering
- LangChain Fundamentals
- AI Agents
- Tool Calling
- LangGraph Basics
- Multi-Agent Collaboration
- Human-in-the-loop Design
- Enterprise AI Workflows